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Engineering Circuit Analysis

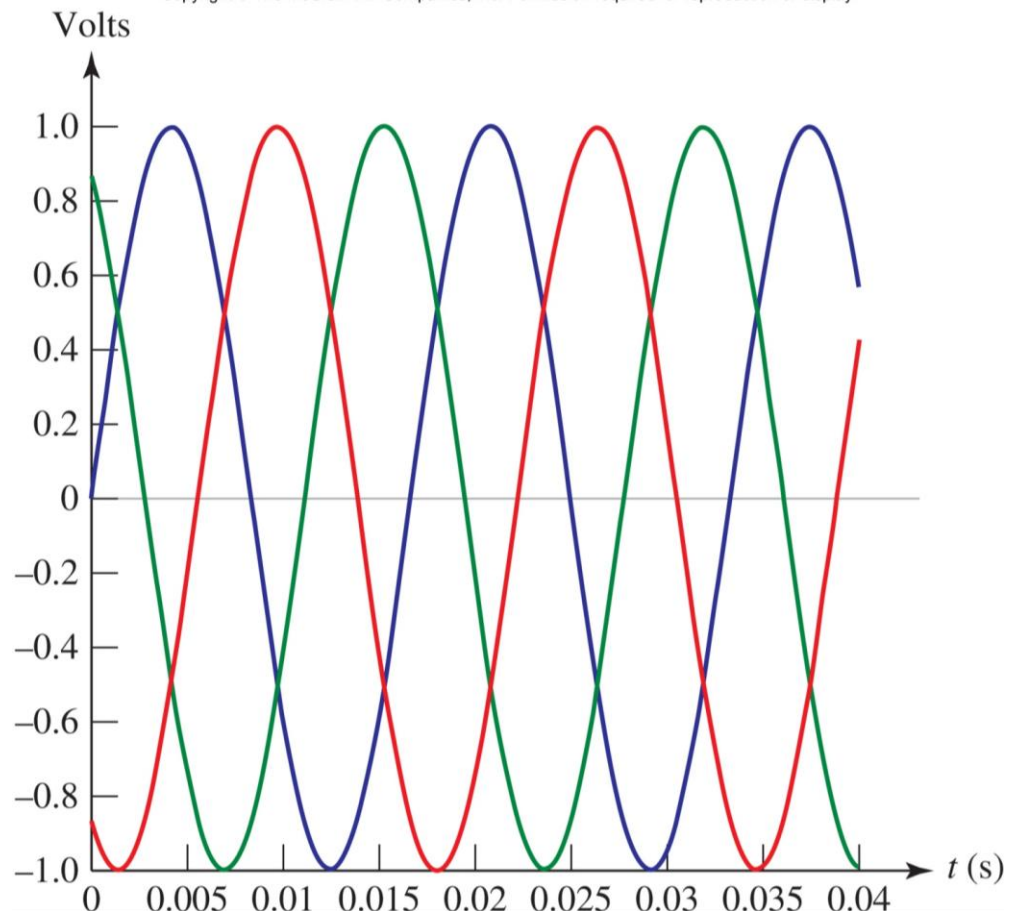
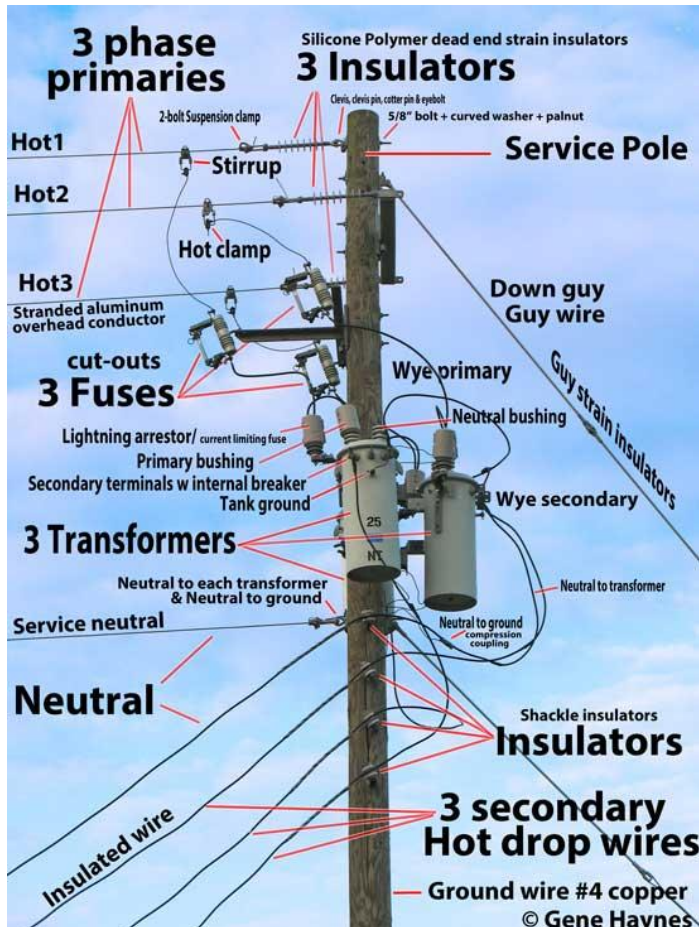
Eighth Edition

Polyphase sources

3-phase systems

- Most industrial motors and machinery systems are wired for 3-phase power
- Easier and cheaper to transmit

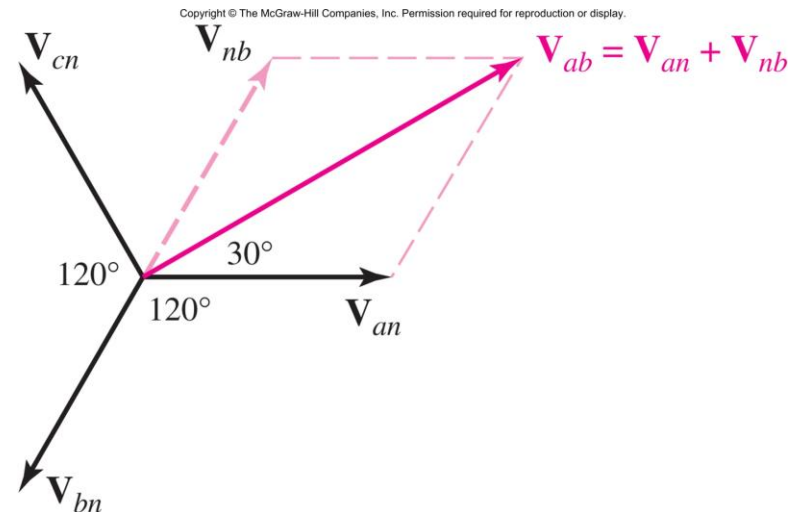
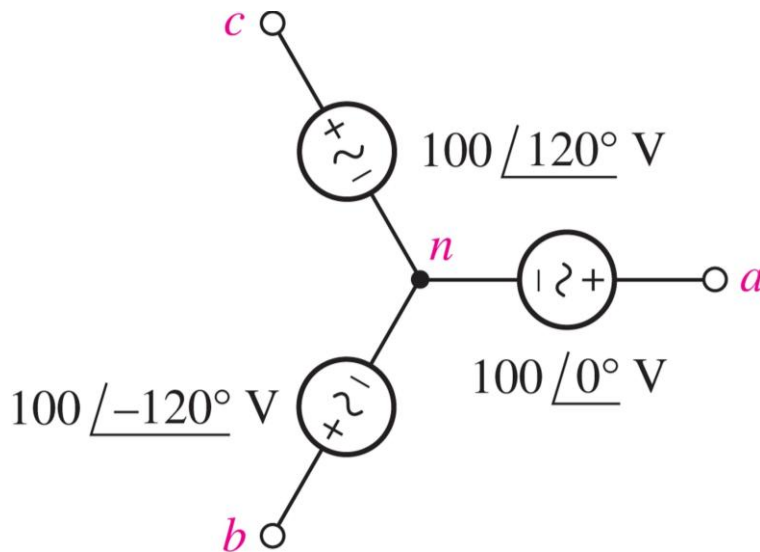
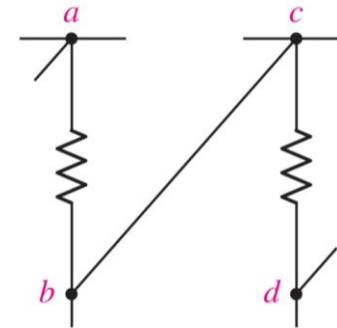
3 wires + ground or neutral wire: between each 2 wires you have sinusoidal voltages of equal amplitude but 120° out-of-phase with each other



3-phase voltage sources

Double script notation:

$$V_{ad} = V_{ab} + V_{bd} = V_{ac} + V_{cd} = V_{ac} - V_{dc}$$



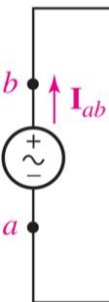
$$V_{an} = 100 \angle 0^\circ \text{ V}$$

$$V_{bn} = 100 \angle -120^\circ \text{ V}$$

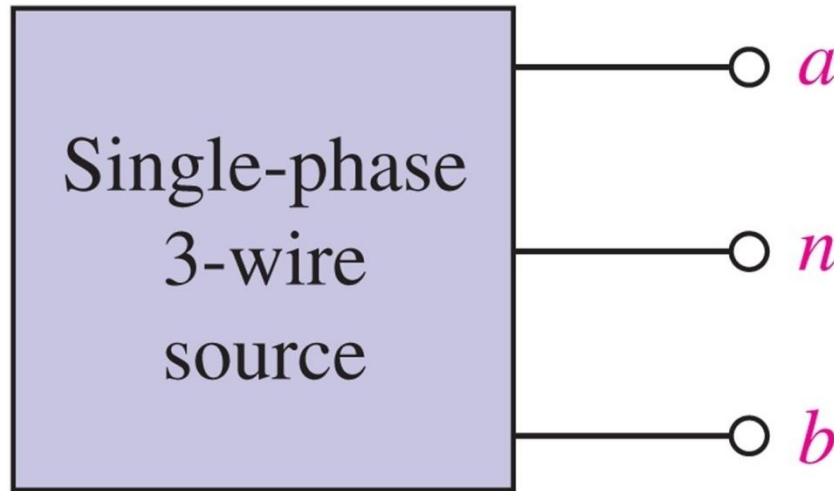
$$V_{cn} = 100 \angle -240^\circ \text{ V}$$

$$V_{ab} = V_{an} + V_{nb} = V_{an} - V_{bn} = 173.2 \angle 30^\circ$$

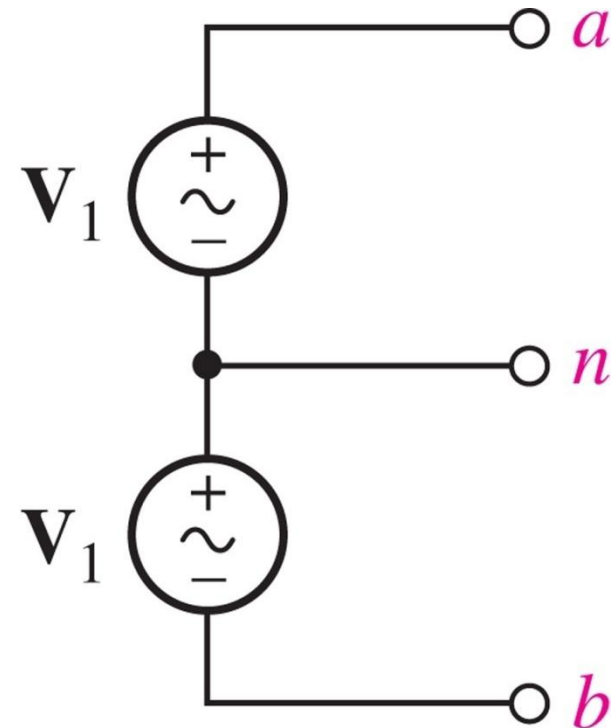
Same script notations are applied to currents



Single-phase 3-wire system



(a)

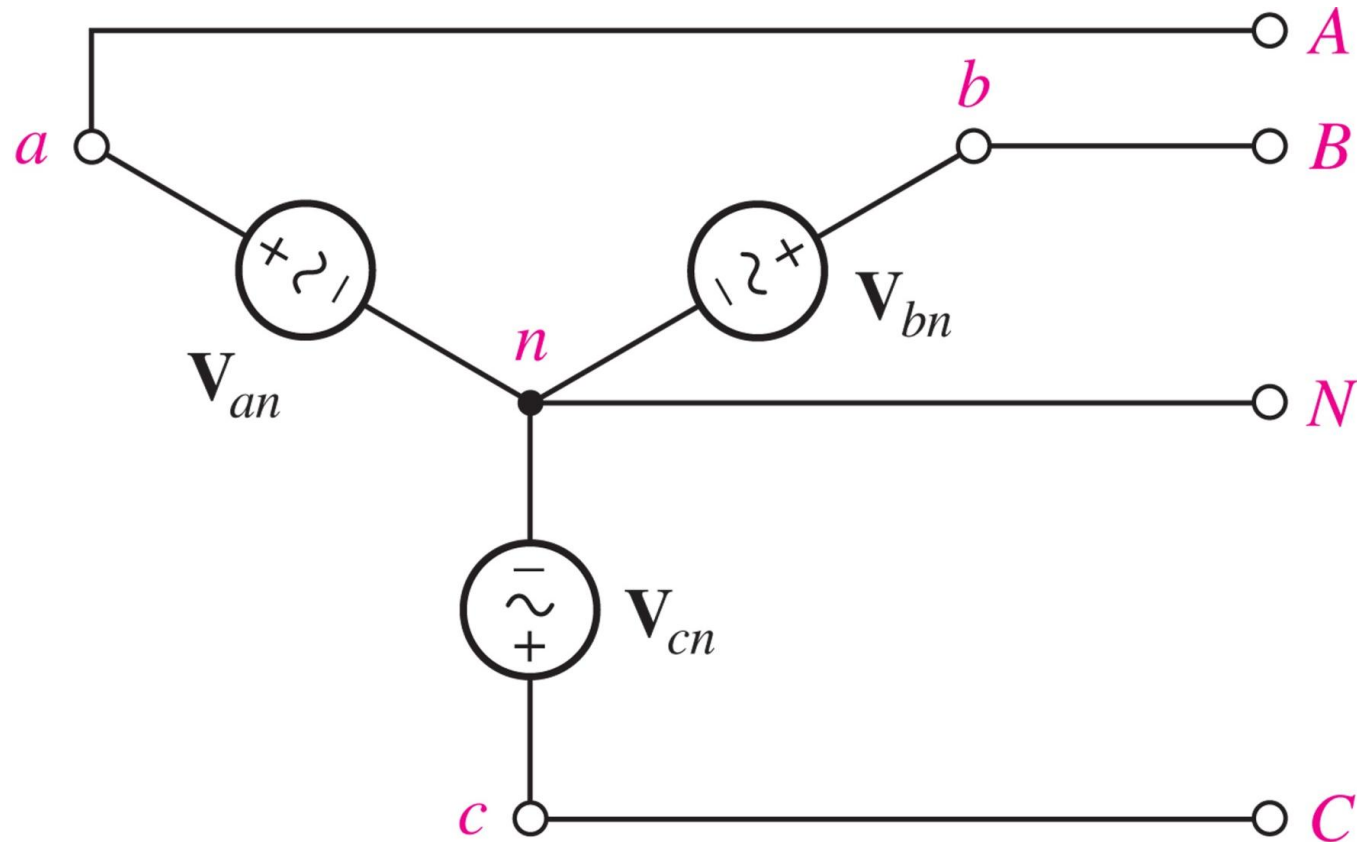


(b)

$V_{an} = V_{nb}$ having the same phase

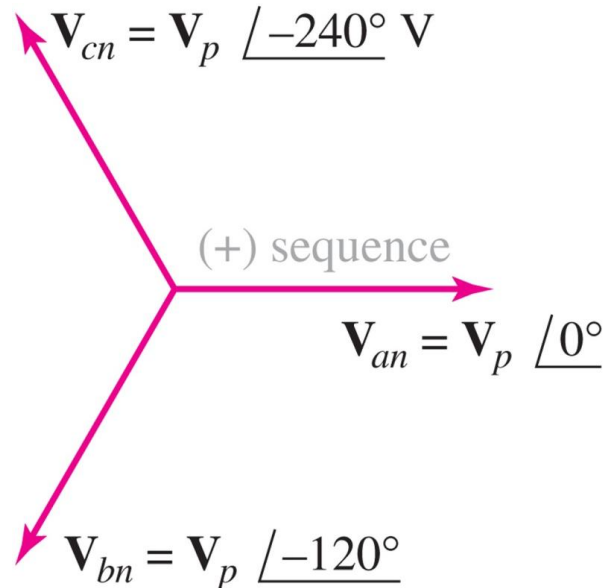
According to the North American standard, it provides 120 V and/or 240 V (all in rms) for US household systems

Three-phase Y-Y (Wye) connection



Balanced sources: $|V_{an}| = |V_{bn}| = |V_{cn}|$
 $V_{an} + V_{bn} + V_{cn} = 0$

Three-phase Y-Y connection



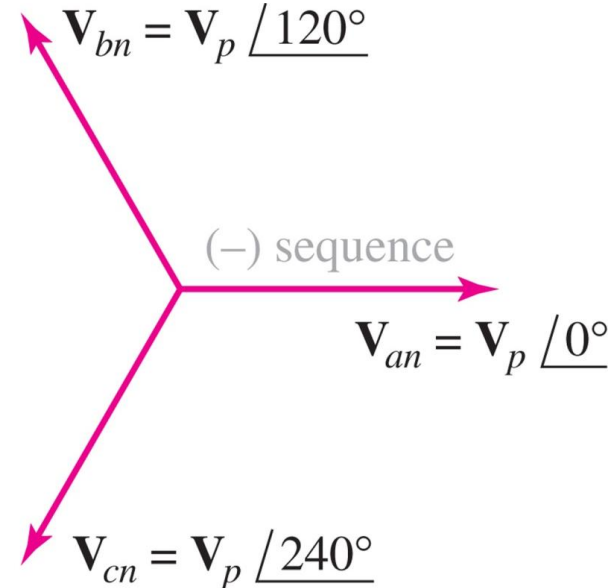
(a)

Positive (abc) phase sequence

$$V_{an} = V_p \angle 0^\circ \text{ V}$$

$$V_{bn} = V_p \angle -120^\circ \text{ V}$$

$$V_{cn} = V_p \angle -240^\circ \text{ V}$$



(b)

Negative (cba) phase sequence

$$V_{an} = V_p \angle 0^\circ \text{ V}$$

$$V_{bn} = V_p \angle 120^\circ \text{ V}$$

$$V_{cn} = V_p \angle 240^\circ \text{ V}$$

Depends on how we choose a-b-c: we assume positive in further considerations

Line-to-Line voltages

Use a phasor diagram

$$V_{ab} = \sqrt{3} V_p \angle 30^\circ \text{ V}$$

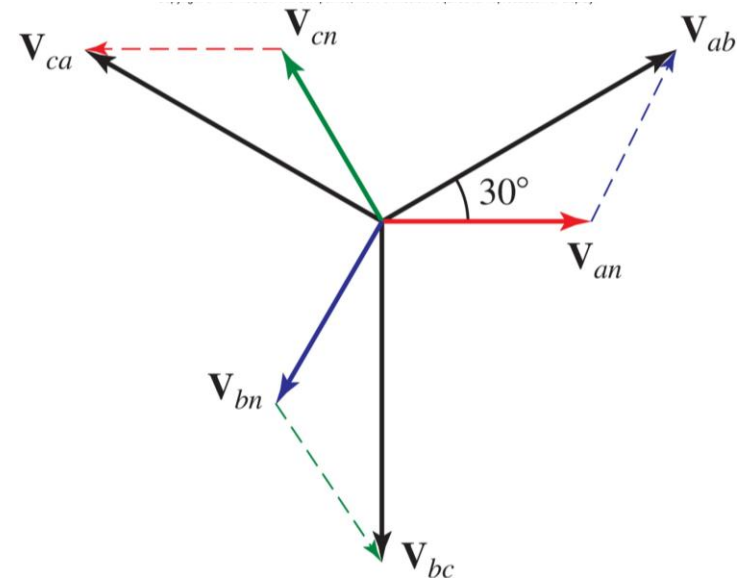
$$V_{bc} = \sqrt{3} V_p \angle -90^\circ \text{ V}$$

$$V_{ca} = \sqrt{3} V_p \angle -210^\circ \text{ V}$$

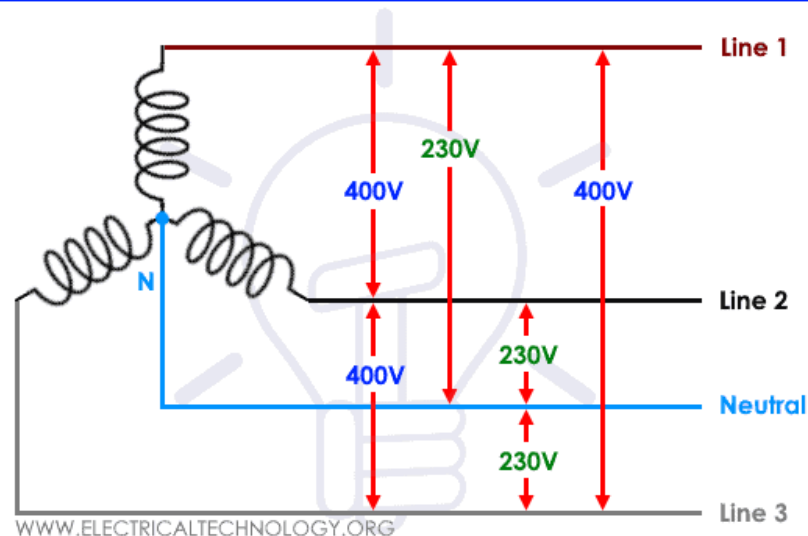


$$V_L = \sqrt{3} V_p$$

$$V_{ab} + V_{bc} + V_{ca} = 0$$



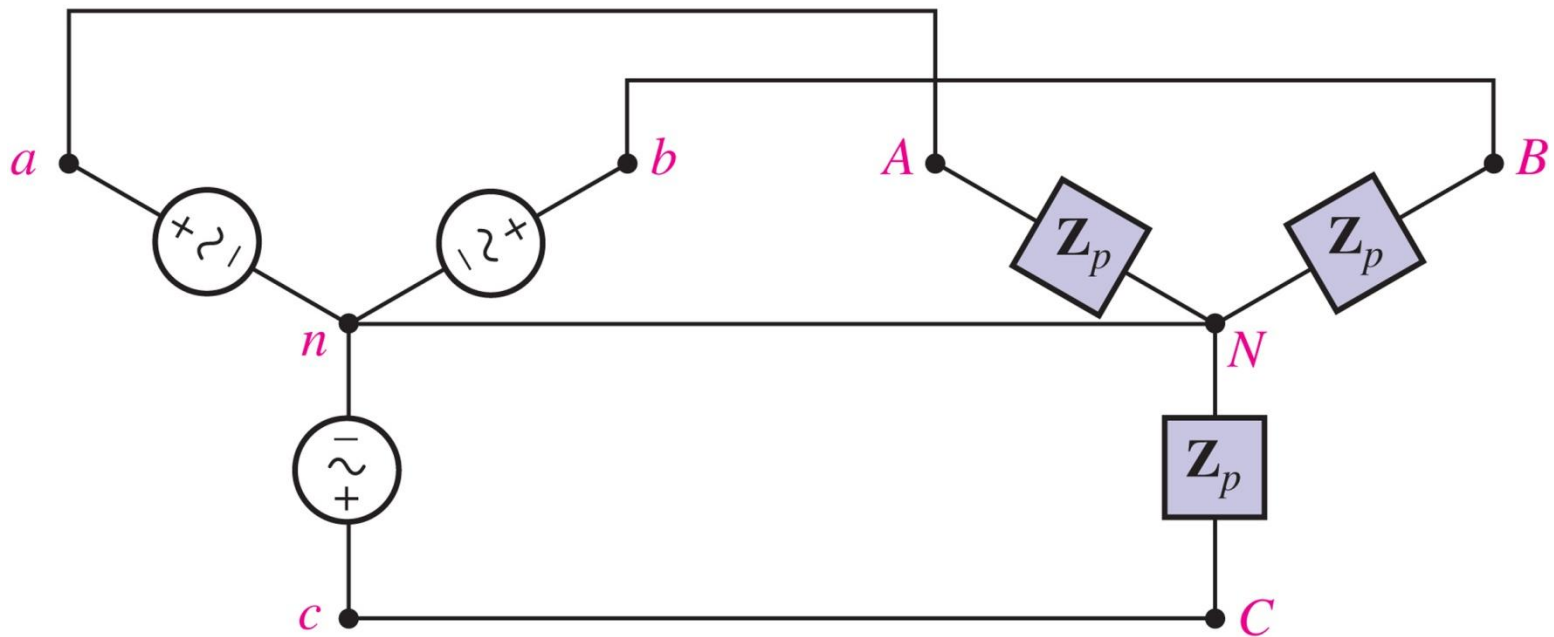
230V & 400V, 1 & 3-Phase Power Supply Systems



Phase to Neutral = 230V, Single Phase
(1-Φ Load can be connected between Any Phase & Neutral)

Phase to Phase = 400V, Three Phase
(3-Φ Load can be connected between All Three Phases)

Three-phase Y-Y connection (balanced load)



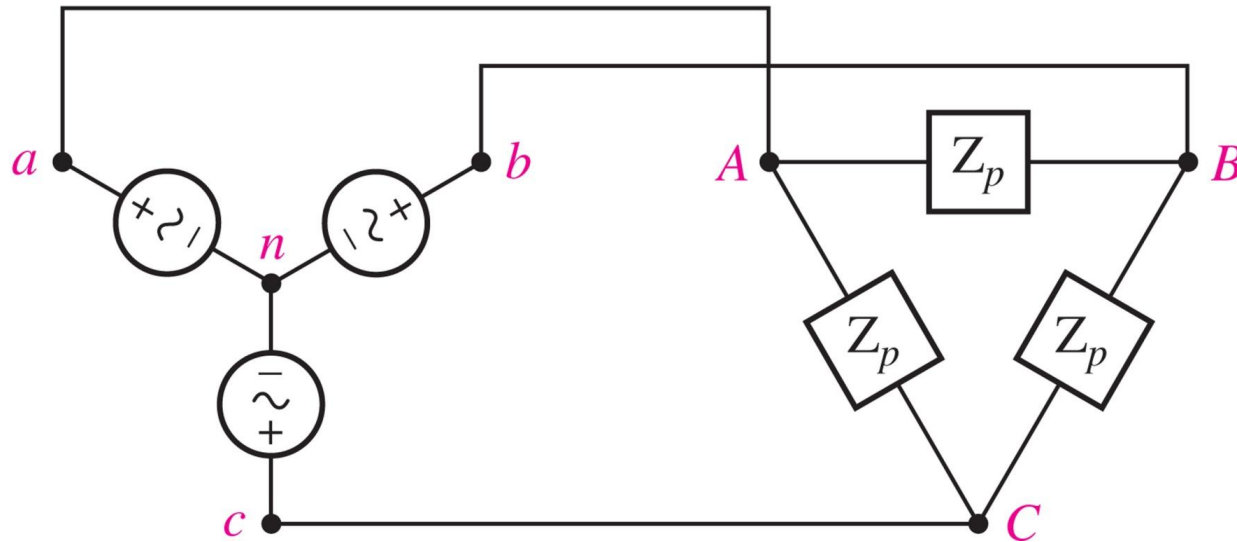
$$I_{aA} = \frac{V_{an}}{Z_p}$$

$$I_{bB} = \frac{V_{bn}}{Z_p} = \frac{V_{an} \angle -120^\circ}{Z_p} = I_{aA} \angle -120^\circ$$

$$I_{cC} = \frac{V_{cn}}{Z_p} = \frac{V_{an} \angle -240^\circ}{Z_p} = I_{aA} \angle -240^\circ$$

$$I_{nN} = I_{aA} + I_{bB} + I_{cC} = 0$$

Three-phase Delta connected load



Very common, doesn't need a neutral wire

For balanced load $V_L = |\mathbf{V}_{ab}| = |\mathbf{V}_{bc}| = |\mathbf{V}_{ca}|$

$$V_L = \sqrt{3} V_p$$

$$V_P = |\mathbf{V}_{an}| = |\mathbf{V}_{bn}| = |\mathbf{V}_{cn}|$$

$$\mathbf{V}_{ab} = \sqrt{3} V_p \angle 30^\circ \text{ V}$$

$$\mathbf{V}_{bc} = \sqrt{3} V_p \angle -90^\circ \text{ V}$$

$$\mathbf{V}_{ca} = \sqrt{3} V_p \angle -210^\circ \text{ V}$$

$$I_{AB} = \frac{V_{ab}}{Z_p} \quad I_{BC} = \frac{V_{bc}}{Z_p} \quad I_{CA} = \frac{V_{ca}}{Z_p} \quad I_{aA} = I_{AB} - I_{CA}$$

$$I_L = |\mathbf{I}_{aA}| = |\mathbf{I}_{bB}| = |\mathbf{I}_{cC}| \quad I_P = |\mathbf{I}_{AB}| = |\mathbf{I}_{BC}| = |\mathbf{I}_{CA}| \quad I_L = \sqrt{3} I_p$$

TABLE 12.1 Comparison of Y- and Δ -Connected Three-Phase Loads. V_p Is the Voltage Magnitude of Each Y-Connected *Source* Phase

Load	Phase Voltage	Line Voltage	Phase Current	Line Current	Power per Phase
Y	$V_{AN} = V_p \angle 0^\circ$ $V_{BN} = V_p \angle -120^\circ$ $V_{CN} = V_p \angle -240^\circ$	$V_{AB} = V_{ab}$ $= (\sqrt{3}/30^\circ) V_{AN}$ $= \sqrt{3} V_p \angle 30^\circ$ $V_{BC} = V_{bc}$ $= (\sqrt{3}/30^\circ) V_{BN}$ $= \sqrt{3} V_p \angle -90^\circ$ $V_{CA} = V_{ca}$ $= (\sqrt{3}/30^\circ) V_{CN}$ $= \sqrt{3} V_p \angle -210^\circ$	$I_{aA} = I_{AN} = \frac{V_{AN}}{Z_p}$ $I_{bB} = I_{BN} = \frac{V_{BN}}{Z_p}$ $I_{cC} = I_{CN} = \frac{V_{CN}}{Z_p}$	$I_{aA} = I_{AN} = \frac{V_{AN}}{Z_p}$ $I_{bB} = I_{BN} = \frac{V_{BN}}{Z_p}$ $I_{cC} = I_{CN} = \frac{V_{CN}}{Z_p}$	$\sqrt{3} V_L I_L \cos \theta$ where $\cos \theta =$ power factor of the load
Δ	$V_{AB} = V_{ab}$ $= \sqrt{3} V_p \angle 30^\circ$ $V_{BC} = V_{bc}$ $= \sqrt{3} V_p \angle -90^\circ$ $V_{CA} = V_{ca}$ $= \sqrt{3} V_p \angle -210^\circ$	$V_{AB} = V_{ab}$ $= \sqrt{3} V_p \angle 30^\circ$ $V_{BC} = V_{bc}$ $= \sqrt{3} V_p \angle -90^\circ$ $V_{CA} = V_{ca}$ $= \sqrt{3} V_p \angle -210^\circ$	$I_{AB} = \frac{V_{AB}}{Z_p}$ $I_{BC} = \frac{V_{BC}}{Z_p}$ $I_{CA} = \frac{V_{CA}}{Z_p}$	$I_{aA} = (\sqrt{3} \angle -30^\circ) \frac{V_{AB}}{Z_p}$ $I_{bB} = (\sqrt{3} \angle -30^\circ) \frac{V_{BC}}{Z_p}$ $I_{cC} = (\sqrt{3} \angle -30^\circ) \frac{V_{CA}}{Z_p}$	$\sqrt{3} V_L I_L \cos \theta$ where $\cos \theta =$ power factor of the load

Commercial buildings have unique electrical requirements that often necessitate the use of 3-phase power and modern distribution systems like power tracks and track sockets.

